



Overriding in Polymorphism:

1. Same name and same parameters are called as Overriding.
2. If the child class is not satisfied with their parent class method implementation, then child class write their own implementation for that method is called as overriding.
3. Overriding is the way to give the additional meaning of existing functionality.
4. We can override methods only.
5. Method overriding happens in parent and child class.
6. We cannot override private, final and static methods.
7. Overriding is also known as “**Run Time Polymorphism**”.
8. It is also known as “**Dynamic Binding**”.

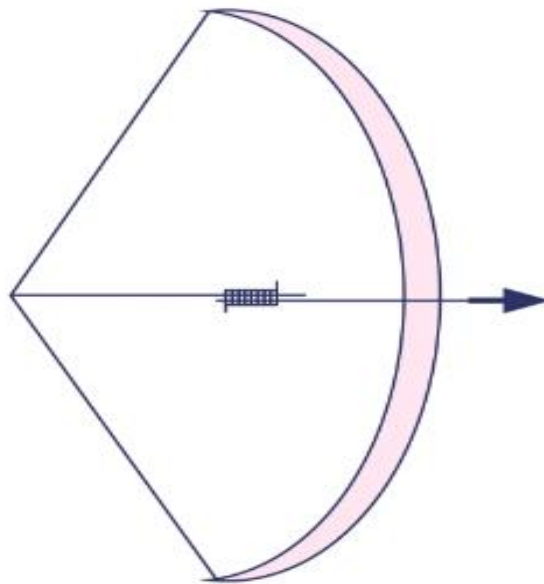


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Overriding



Overriding Diagram

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Syntax rules for the Overriding:

- Method name must be same.
- Method parameter must be same.
- Method overriding only happens in the parent and child class.
- Method return type is void or primitive then it must be same.



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- Method return type is class type then it must be same or subclass (co-variant return type).
- Access modifier should be same or greater but it cannot be weaker.

public > protected > default > private

- In java 1.5/ jdk 1.5 **@Override** annotation is added so that compiler confirms that the method is overridden or not.

What is Run Time Polymorphism?

- This type of polymorphism in java is also called dynamic binding.
- It can be achieved by method overriding.
- In this process, an override method call is resolved at run time rather than resolving at compile time.

Understanding Program:

```
public class A  
{
```



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```
public void m1()
{
    System.out.println("m1---A");
}
}
```

```
public class B
{
    public void m1()
    {
        System.out.println("m1---B");
    }
}
```

```
public class Test
{
    public static void main(String args[])
    {
        A a = new B();
        a.m1();
    }
}
```



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Can We Override Static Methods?

No, we cannot override static methods.

Reason – Method overriding relies on dynamic binding at runtime, but static methods are bonded at compile time with static binding. As a result, we are unable to override static methods.

Static methods also known as class level methods.

Can We Override Final Methods?

No, we cannot override final methods.

Reason – When we want that the content of a method should not change by any outsider or child class, then we declare the method as the final method in java.

When a method is declared as the final method in the parent class, then any child class cannot override or modify the final method in java.



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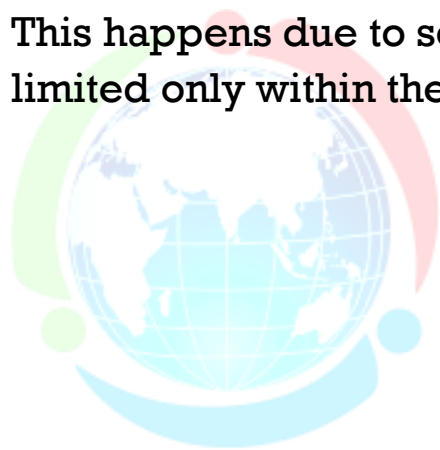
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Can We Override Private Methods?

No, we cannot override private methods.

Reason – Private methods are not visible to the child class; they are only visible and accessible in the class on which they are declared.

This happens due to scope of private access modifier is limited only within the class.



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